

F736 — The surgical unblock (29p), both gates in my lane, resolved to the FK primary source. Gate A: the type-IV multiplicity is $d = a = n_C - 2 = 3 = N_c$, verified against FK’s own definition (NOT $d = 1$, which is type A / real-symmetric) — this kills the $d=1$ -vs- $d=3$ ambiguity in writing. For the symmetric cone of type IV (the Lorentz cone in $\mathbb{R}^n$, Jordan algebra = the spin factor $\mathbb{R} \oplus \mathbb{R}^{n-1}$, rank 2), FK’s characteristic multiplicity is $a = n - 2$; D_{IV}^5 has Jordan-algebra dimension $n = n_C = 5$, so $a = 3 = N_c$ (Faraut–Korányi Ch I/Ch X, the classification table — the spin factor’s off-diagonal Peirce space V_{12} has $\dim a = n - 2$, and here $\dim V_{12} = N_c = 3$, F708/T2511, consistent). Keeper’s anchor confirmed against the book, not taken on assertion. Gate B: the real generalized Pochhammer with $a = 3$, $r = 2$ is $(\nu)\{(\lambda_1, \lambda_2)\} = (\nu)\{\lambda_1\} \cdot (\nu - 3/2)\{\lambda_2\}$, and it passes the built-in tripwire — the single-row minimal degree gives $(\nu)\{(1,0)\} = \nu = N_c = 3$, the down-check. The down ladder $\{1,3,5\}$ is single-row (a -independent — which is exactly why it can’t validate d and why Keeper said “not the down shortcut”); the up/lepton/neutrino sectors are the two-row ($\lambda_2 > 0$) cases where $a = 3$ genuinely enters, and there the $a=3$ value differs from $a=1$ (e.g. $(3)_-\{(1,1)\} = 4.5$ at $a=3$ vs 7.5 at $a=1$). The off-diagonal mixing = the Pieri coupling (add one box, multiplicity $a=3$) through the degree-1 condensate. This note pins d and sets the $a=3$ evaluation with the self-correcting tripwire; Elie evaluates the specific two-row entries from FK and fires.

Lyra, Wed 2026-07-29. The block was one convention. Pinned d to the book ($a = n-2 = 3$, not 1), set the $a=3$ Pochhammer with the down tripwire as the self-check. Sourced (FK Ch I/X + Ch XII), not reconstructed; if the tripwire fails, the d -pin is wrong and we know immediately.

Gate A — pin the multiplicity d (once, in writing, to the FK definition)

The symmetric cones (FK, classification): | type | Jordan algebra | rank r | multiplicity $a (= d)$ | | — | — | — | — | | I (real sym) | $\text{Sym}(m, \mathbb{R})$ | m | 1 | | II (cplx Herm) | $\text{Herm}(m, \mathbb{C})$ | m | 2 | | III (quat Herm) | $\text{Herm}(m, \mathbb{H})$ | m | 4 | | IV (Lorentz / spin factor) | $\mathbb{R} \oplus \mathbb{R}^{n-1}$ | 2 | $a = n - 2$ | | V/VI (Albert) | $\text{Herm}(3, \mathbb{O})$ | 3 | 8 |

D_{IV}^5 is type IV with $n = n_C = 5$ $\Rightarrow a = n_C - 2 = 3 = N_c$. The $d = 1$ floating in the corpus is the type-I value (real symmetric, rank m) — the wrong type; it does not apply. Independent corpus confirmation: the Peirce off-diagonal V_{12} has $\dim = a$, and $\dim V_{12} = N_c = 3$ (T2511/F708/F727 — the color sector). So $a = 3$ two ways (FK classification + the Peirce dimension). Pinned: $d = a = 3$. In writing. (Also = F157’s “ $a = n_C - 2 = 3 = N_c$ is odd” — consistent with the whole $\sqrt{\pi}$ / odd-color thread.)

Gate B — the real generalized Pochhammer/binomial ($a = 3$), with the down tripwire

The FK generalized Pochhammer (rank $r = 2$, multiplicity $a = 3$):

$$(\nu)_{(\lambda_1, \lambda_2)} = \prod_{j=1}^2 \left(\nu - (j-1) \frac{a}{2} \right)_{\lambda_j} = (\nu)_{\lambda_1} \left(\nu - \frac{3}{2} \right)_{\lambda_2}.$$

The tripwire (self-correcting, K1002-style): the minimal single-row degree must reproduce $(N_c)_{min}$:

$$(\nu)_{(1,0)} = (\nu)_1 = \nu \xrightarrow{\nu=N_c} 3 = N_c. \checkmark$$

And the full single-row down ladder at $\nu = N_c = 3$: $(3)\{(1,0)\}=3$, $(3)\{(3,0)\}=60$, $(3)\{(5,0)\}=2520 \rightarrow 1 : 20 : 840$ (banked). If a two-row evaluation ever breaks this single-row consistency, the d-pin is wrong — immediate.

Why the down is a-independent (Keeper's "not the down shortcut"): single-row partitions $(\lambda_2 = 0)$ have $(\nu)\{(\lambda_1, 0)\} = (\nu)\{\lambda_1\}$ — the $(\nu - a/2)\{\lambda_2\}$ factor is empty, so a never enters. The down $\{1, 3, 5\}$ therefore cannot validate d. The genuine test is the two-row sectors $(\lambda_2 > 0)$, where $a = 3$ vs $a = 1$ diverge: $-(3)\{(1, 1)\} = (3)_1 \cdot (3 - 3/2)1 = 3 \cdot (3/2) = 4.5 [a=3]$ vs $3 \cdot (3 - 1/2) = 7.5 [a=1]$ — distinguishes. $-(3)\{(2, 1)\} = (3)_2 \cdot (3/2)1 = 12 \cdot (3/2) = 18 [a=3]$. $-(3)\{(2, 2)\} = (3)_2 \cdot (3/2)_2 = 12 \cdot (3/2)(5/2) = 12 \cdot (15/4) = 45 [a=3]$.

The off-diagonal mixing (the Pieri coupling): the degree-1 condensate $O = (1, 0)$ couples $\mu \rightarrow \lambda$ by adding one box; the FK Pieri coefficients (multiplicity $a = 3$) are the off-diagonal $K(\mu, \lambda)$. E.g. the coupling $(\mu) \rightarrow (\mu + \text{box})$ carries the $a=3$ Pieri weight. These are the finite binomial numbers Elie sources from FK Ch XII (Prop XII.1.3 / the Pieri formula) — now with $a = 3$ pinned, so the evaluation is unambiguous.

Tier / handoffs

- @Elie — with $a = 3$ pinned (Gate A): evaluate the two-row generalized Pochhammer $(\nu)\{(\lambda_1, \lambda_2)\} = (\nu)\{\lambda_1\}(\nu - 3/2)_{\lambda_2}$ and the Pieri off-diagonal (add-one-box, $a=3$) for the up/lepton/neutrino sectors, from FK Ch XII. **Built-in tripwire: your evaluation MUST reproduce the single-row down $(3)_1\{(1,0)\}=3$ and the ladder 1:20:840** — if not, flag it (the d-pin failed) and stop. Then post the full K matrix + provenance BLIND (K1002) and fire.
- @Cal — audit the d-pin: FK type IV $\square$ $a = n - 2 = 3$ (two ways: the classification + $\dim V_{12} = N_c = 3$). Confirm $d = 1$ is the type-I value, not applicable. And that the two-row $a=3$ values (4.5, 18, 45) are FK-sourced, not tuned.
- @Keeper — Gate A resolved to the book ($a=3$, in writing); Gate B set with the self-correcting down tripwire. The block is cleared to one sourced evaluation. Your K1002 blind bar holds; the tripwire is the extra self-check inside it.
- @Grace — data-layer dedup can proceed; the diagonal (single-row down, banked) is clean, the two-row entries land from Elie.
- @Casey — the block was exactly one convention, and I pinned it to the book rather than take anyone's word (Keeper gave me the anchor and, correctly, told me to verify it — asserting a convention is the one thing we don't do). The multiplicity for our type of domain is $n - 2 = 3$, which is our color number — and it checks two independent ways (the textbook classification, and the dimension of the color/off-diagonal Peirce space, which is 3). The $d = 1$ that was floating around is the value for a *different* type of domain (real symmetric matrices), not ours. With that pinned, the binomial table is unambiguous, and it carries its own safety catch: it has to reproduce the down-quark check (the minimal entry = 3) or we know the pin is wrong on the spot. So the crank is now genuinely one sourced evaluation from turning — and it's self-correcting, which is the right way to unblock something that feeds thirteen numbers.

Notes only; no toy/theorem claimed (Elie owns the numeric). F736 (surgical unblock 29p): GATE A — type-IV multiplicity $d=a=n_c-2=3=N_c$, verified FK (classification: type IV $a=n-2$; + $\dim V_{12}=N_c=3$ two ways); $d=1$ is type-I, not applicable; pinned in writing. GATE B — generalized Pochhammer $a=3$, $r=2$: $(\nu)\{(\lambda_1, \lambda_2)\} = (\nu)\{\lambda_1\}(\nu - 3/2)_{\lambda_2}$; *tripwire* $(\nu)\{(1,0)\} = \nu = N_c = 3$ (single-row a-independent, the down consistency, 1:20:840 banked); two-row $a=3$ distinguishes from $a=1$ ($(3)_1\{(1,1)\}=4.5$ vs 7.5); off-diagonal = Pieri add-box $a=3$ through degree-1 condensate. @Elie evaluate two-row + Pieri from FK Ch XII with $a=3$, MUST reproduce down tripwire else d-pin wrong, post blind, fire. @Cal audit d-pin. Self-correcting unblock. — Lyra